Aangan — Install Guide & Idea Menu

Nathaniel School of Music · 29 Jul 2026 · Part 1: House Pi from scratch · Part 2: the two door screens · Part 3: Music Pi pointer · Part 4: every creative idea, studio + music, with what it becomes in the app

Status: the Aangan app is LIVE at **studio-command.vercel.app** with the state dial, Pre-flight, Displays + delivery OTP, Family SOS (**/#/sos**) and the close-the-door nudge — running in simulation until the House Pi below goes live. Repo: **github.com/jasonzacmusic/aangan**.

PART 1 — House Pi from scratch
Home Assistant OS · ~2 hours hands-on, sensors over a weekend

1 **Flash Home Assistant OS.**

On the Mac: install **Raspberry Pi Imager** → CHOOSE OS → *Other specific-purpose OS* → *Home assistants and home automation* → *Home Assistant* → pick Raspberry Pi 5 → write to a 64GB A2 card. No settings needed — HA configures itself.

2 **First boot.**

Card into the new Pi 5, Ethernet cable to the router if possible, power with the official 27W PSU. Wait ~5 minutes, then open `http://homeassistant.local:8123` on the Mac. Create the admin account, name the home **Aangan**, set location Bangalore.

3 **Install the ESPHome add-on.**

Settings → Add-ons → Add-on store → **ESPHome Device Builder** → Install → Start (turn on “Show in sidebar”).

4 **Flash the six ESP32 zone nodes.**

The six YAML files live in the repo at `pi/house/esphome/`. Copy `secrets.example.yaml` → `secrets.yaml` (Wi-Fi name/password + one generated key per node). First time: plug each ESP32 into the Mac by USB and flash from the ESPHome page; after that all updates are over Wi-Fi. Adopt each node under Settings → Devices & Services.

5 **Install the Aangan package.**

Copy `pi/house/homeassistant/packages/studio_command.yaml` to `/config/packages/` (use the File editor add-on), add the `packages:` include to `configuration.yaml` as written in `pi/house/README.md`, fix any entity names that adopted differently, restart HA. This creates **binary_sensor.studio_ready** — the one authoritative go/no-go verdict.

6 **Family phones.**

Everyone (Amma, Jason, brother) installs the **Home Assistant Companion** app and logs into `http://homeassistant.local:8123` on the home Wi-Fi. iPhones: when asked, ALLOW **Critical Alerts** — that is what makes SOS ring through silent mode. Add each phone to the `notify:` group in the package.

7 **The wrapper (feeds the app).**

Run `pi/house/wrapper/studio_wrapper.py` on any always-on LAN machine (the Piano Pi is fine) on port **8126**, with a long-lived HA token. Then in the app repo set `USE_MOCK = false` and the wrapper URL in `src/config.ts`. Claude does this step.

The wrapper now also implements the **/api/sos** endpoints (contract in the repo README) so the Family SOS is real: hold on any phone → HA critical alerts + signs flash + panels take over.

8 **Sensors, safety-first order.**

⓪ Counts are settled from the WhatsApp inventory: 0 reeds and 1 leak sensor owned — the procurement list orders 12 reeds and 9 leak sensors. ① Certified smoke + LPG alarms mounted and tested standalone FIRST. ② Door reeds on the 4 double doors — at the **meeting stiles (middle)**: switch on one leaf, magnet on the other, so either leaf opening trips it. ③ MQ-6 near each stove, flame sensors, leak probes at every sink/stove line, DHT22 in each bedroom. Each ESP32 node gets its own 5V wall charger (on the buy list). ④ The panic-loop and any mains wiring is ONLY done by the low-voltage/fire-alarm technician; ESP32s just read volt-free contacts. The owned buzzers, soil-moisture sensor and HC-SR04 (overhead tank) wire into the same nodes.

8b **Doorbell camera v1 — ₹0.**

The owned ESP32-CAM mounts at the main-entrance door frame (its assigned job in the sourcing matrix) with a momentary push button as the bell. Snapshots already flow to the app's Safety page and the panels. The Tapo C210 on the procurement list is the proper entrance camera; the ESP32-CAM stays as the door-frame secondary.

9 **Commission.**

Run every line of `docs/TEST_CHECKLIST.md`: trip each sensor for real, pull a node's power and watch studio_ready hold, test SOS from Amma's phone, test a delivery OTP on the door screen. Not “done” until every box ticks.

PART 2 — The two door screens

Android tablets in kiosk mode — no Pi Zero, one cable each · ~20 min per screen

1 **Set up each tablet once.**

Sign in with the school Google account, install **Fully Kiosk Browser** from the Play Store, set its Start URL to `https://studio-command.vercel.app/#/display/front-house` (front door) or

`/#/display/front-studio` (studio door), and turn on: kiosk mode, screen always on, auto-start on boot, brightness by daytime.

2 **Mount.**

Screw the wall mount beside the door, clip the tablet in, run the flat USB-C cable down the wall to the charger. The tablet's own battery rides through power blips, so the door sign never goes black mid-take.

3 **Assign in the app.**

Displays page → front door = “Door sign” (COME ON IN / RECORDING + the delivery OTP takeover + the guest QR); studio door = “Studio state” (ON AIR + the close-the-door nudge). The old iPad remains the third, in-house panel. Once the House Pi is live, the same app served on the LAN keeps panels working even when internet is down.

PART 3 — Music Pi built in PARALLEL with the house · separate order & invoice

The complete order-to-first-note build (Raspberry Pi DAC Pro + XLR board + Pianoteq Pro, every connection, every command, 16-point test) is its own document: **PIANO_PI_BUILD_PLAN.html** in the repo's `docs/`. Both projects run together: while robu ships the sensors and boards, the House Pi gets Home Assistant (Part 1) and the owned Pi 5 gets its OS flashed and Pianoteq activated — the DAC Pro just slots on when it arrives. One correction Claude will make at build time: the repo's setup script still writes the old HiFiBerry overlay — it changes to the DAC Pro (`iqaudio-dacplus`) family.

PART 4 — The Idea Menu everything the two Pis (+ Apple TV) can become · tick what you want, Claude builds it into the app

Studio & House

IDEA	WHAT IT DOES	NEEDS	IN THE APP	STATUS
Family SOS	One hold on any phone raises the whole house; wall panels show who needs help	nothing new	/#/sos + Safety page	SHIPPED
Delivery OTP + directions	2-tap hand-off; OTP huge on the door screen with standing directions	door screens	Displays page	SHIPPED
Close-the-door nudge	Door open + room loud → panels and phones ask for the door	door reeds + dB meter	banner everywhere	SHIPPED
Apple TV = HomeKit hub	HA's HomeKit Bridge exposes Aangan to iOS: “Hey Siri, is the studio ready?”, states as scenes in the Home app, family presence automations — the	Apple TV (owned)	works alongside the app	BUILT · pairs at install

	Apple TV is the always-home hub that makes Siri work from anywhere			
Calendar-driven states	Class slots in Google Calendar auto-set the dial to Class and back — no human forgets the sign	nothing	Command page automation	BUILT · pairs at install
Tarang on the studio TV	MIDI-reactive visuals (the existing Tarang engine) on a wall TV during classes and tours; the dial's Class state turns it on	HDMI from a Pi to the TV	a “Visuals” toggle on Command	APPROVED
Fleet health card	One card answers “is everything in the school alive?” — Macs, Pis, printers, router (nsm-health feeds it)	nothing	new Home card	APPROVED
Voice paging	“Class starting in 5” spoken/shown on the wall panels from the app	panel speakers optional	Displays page button	APPROVED
Guest QR at the door	Visitor scans a QR on the door screen → their phone shows live wait status + “come in softly” etiquette	nothing	door panel + tiny page	APPROVED
Energy monitor	Per-circuit power on House Pulse — what the studio actually costs per month, AC left-on alerts	CT-clamp energy meter (electrician)	House Pulse	IDEA
Doorbell cam v2	Reolink 2K doorbell (on the buy list — go-for-broke) with person detection via Frigate — snapshot in history + on phones	Reolink doorbell (ordered)	Safety page	APPROVED
Laundry / fridge nudges	Spare SW-420 says “washing machine done”; reed + DHT22 says “fridge door open”	owned spares	history + notifications	APPROVED
Rain-on-the-balcony alert	Owned leak sensor outside → “bring the clothes in” on every panel	owned spare	notification	APPROVED

Music rig

> **Honest note:** the DAC Pro is output-only (that's why it sounds so good for its job). So music-Pi ideas are MIDI + playback ideas — anything needing audio *inputs* stays on the Mac/console side.

IDEA	WHAT IT DOES	NEEDS	IN THE APP	STATUS
MIDI black-box	The rig silently records EVERY note ever played as MIDI (tiny files, 24/7). An improvisation is	nothing new	Piano Rig card: “last	SHIPPED

	never lost — replay any moment through Pianoteq, or export it into a lesson		takes" list + replay cue	
Practice stats	The black-box becomes numbers: minutes played per day, streaks, most-practised keys/tempo — for Jason AND students	black-box	Piano Rig card graph	IDEA
Preset picker	Full Pianoteq instrument list in the app (not just next/prev) — a NY Steinway for class, a Rhody for the evening, from any phone	nothing	Piano Rig card	NEXT
NSM sample player	sfizz on the rig plays OUR libraries (Salamander + any SFZ we build on the Mac) beside Pianoteq — tanpura drones, pads, harmonium	nothing new	Piano Rig: sound source switch	LATER
Tanpura + metronome on the rig	Drone in any key + the Groove Metronome on the rig's touch screen — the practice companion lives where the piano is	nothing	rig touch UI	IDEA
Layer/split mode	Piano + pad layered, or bass/piano split — one keyboard, a fuller stage sound. **MIDI in every form:** USB native, 5-pin DIN via Roland UM-ONE, wireless via Bluetooth MIDI + CME WIDI Master (both on the buy list)	UM-ONE + WIDI (ordered)	rig touch UI	APPROVED
AirPlay into the console	The rig is an AirPlay/Spotify Connect receiver — any phone plays reference tracks through the XLR outs	nothing	Piano Rig toggle	IDEA
REAPER remote	The rig's touch screen gets a Mac mode: transport, record-arm, marker drop — control the recording Mac from the piano bench	nothing	rig touch UI	IDEA
Nightly REAPER backup	3am rsync of the Mac's REAPER projects to the rig's USB SSD — an off-Mac copy always exists	USB SSD (already owned)	Fleet card shows "last backup"	APPROVED
Recording tally	Rec states already cue the rig; the rig screen shows a red ON AIR tally at the piano	nothing	already wired	SHIPPED

How approval works: tell Claude which rows to build ("build the MIDI black-box and the Apple TV HomeKit bridge") — SHIPPED means live today, NEXT means ready to build immediately, IDEA means a short build after the hardware lands.